

SATQUERY AI — REMOTE SENSING MISSION REPORT

Indian Space Research Organisation (ISRO) | SIH 26167 Intelligence Dispatch

Natural Language Query:	Detect urban expansion and land use change between 2022 and 2024
Identified Workflow Intent:	CHANGE_DETECTION
Dispatched Specialist:	ChangeDetectionEngine
Composite Confidence:	93.7% (HIGH)

1. Executive Analysis & Findings

Assessment: Bi-temporal spectral difference analysis detected 10.7% spatial modification across 2.79 km² of the observed scene. Impervious built-up surfaces shifted by +9.5%, while vegetative cover shifted by -9.3%. Spatial co-registration confirmed 4 localized change cluster(s).

- Total changed surface: 2.79 km² (10.7% of raster footprint)
- Impervious built-up change: +9.5%
- Vegetation canopy change: -9.3%
- Detected 4 discrete spatial change cluster(s)

2. 4-Signal Multi-Criteria Confidence Matrix

Metric Signal	Component	Score (%)	Status
C_model	Specialist Model Inference Certainty	91.0%	Optimal
C_sensor	Sensor Radiometric SNR & Quality	90.0%	Optimal
C_alignment	Spatial Co-Registration (CRS & IoU)	100.0%	Verified
C_resolution	Ground Sample Distance Suitability	95.0%	Optimal

3. Auditable Agent Execution Pipeline Trace

Step	Action / Milestone	Tool Executed	Latency	Status
1	Raster Ingestion & Metadata Parsing	GeoTIFFProcessor	0.9 ms	COMPLETED
2	Spatial Co-Registration & Alignment Vali	SpatialRegistrationValidator	0.6 ms	COMPLETED
3	Agentic Query Classification & Tool Rout	AgentClassifier	0.0 ms	COMPLETED
4	Specialist Model Execution (ChangeDetect	ChangeDetectionEngine	55.6 ms	COMPLETED
5	Evidence Fusion & 4-Signal Confidence Ra	CompositeConfidenceEngine	0.0 ms	COMPLETED